

The Sparsity Ceiling: Where Spiking Networks Can—and Cannot—Trade Activity for Energy

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Abstract

Spiking neural networks (SNNs) are promoted as an energy-efficient substrate because sparse, event-driven activity replaces dense multiply-accumulates (MACs) with cheap accumulates (ACs). We argue that the energy dividend of sparsity is *not a property of SNNs but of the task*. Using a controlled protocol—architecture held fixed, only the hidden unit swapped (continuous vs. leaky-integrate-and-fire), plus a two-sided target-firing-rate probe—we measure how far activity can be pushed down before quality breaks. On feed-forward perception, hidden firing is naturally low and drives precisely to a 5% target with no accuracy loss. On recurrent character-level language modeling the same probe, targeting 10%, cannot move firing below $\sim 50\%$: the recurrent state must stay active to carry information. A spiking *Transformer*, by contrast, sparsifies freely to 2% firing at no extra quality cost (3 seeds) — so the ceiling is a property of *recurrent compression*, not of sequence modeling. Attention escapes the floor only by storing the full key-value cache, trading a firing floor for a memory wall: neither architecture escapes cost on neuromorphic hardware, they pay on different axes. We formalize the ceiling with an information-theoretic firing-floor bound $\rho \geq H_b^{-1}(\log_2 M/H)$ and confirm its predictions on synthetic tasks: the floor rises with memory load and falls with state width, and—refuting a naive memory-only reading—also rises with task difficulty, exactly as a load-based bound requires. A layer-wise accounting exposes an *input floor* that further caps op reduction under dense (frame-replayed) input, isolating native event-driven input as the setting where neuromorphic hardware wins. The results recast “SNNs save energy” as a task-conditional claim with a concrete, measurable rule for where neuromorphic computation pays off.

1 Introduction

The standard case for neuromorphic computation is energetic: spikes are binary and sparse, so inference reduces to accumulate (AC) operations gated by activity, avoiding the dense multiply-accumulates (MAC) of conventional networks. This motivates a large literature reporting 10–100× energy reductions and, increasingly, spiking

language models [1, 2, 3].

We ask a narrower, falsifiable question: *when a network is otherwise identical, how much energy does spiking actually buy, and what determines that number?* Prior work overwhelmingly optimizes accuracy on a single task and reports an energy estimate at whatever firing rate results; it rarely isolates the causal variable—attainable sparsity—or asks whether that variable is controllable in a given task. Our contributions:

- **Protocol.** A matched-architecture comparison changing only the neuron model, plus a two-sided target-rate regularizer that *probes* attainable sparsity rather than accepting the incidental one.
- **The sparsity ceiling.** Low-load feed-forward perception is freely sparsifiable (to $\sim 5\%$ firing); recurrent sequence modeling is not (firing pinned near 50–63%). Sparsity availability is task-structural.
- **Theory + consequence.** An information-theoretic firing-floor bound $\rho \geq H_b^{-1}(\log_2 M/H)$ explains the ceiling; controlled sweeps confirm the floor rises with memory load and falls with state width, and (refuting a naive memory-only reading) also rises with task difficulty—the floor tracks representational load $\log_2 M$, which memory blows up exponentially. A layer-wise *input floor* shows dense-input replay caps op reduction.
- **Recurrence vs. attention: a cost dichotomy.** A spiking Transformer sparsifies freely to 2% (no floor), localizing the ceiling to recurrent compression; attention pays instead in key-value memory. The practical rule: neuromorphic wins on *event-driven perception*; on sequence models it faces a firing floor (recurrent) or a memory wall (attention).

2 Method

Matched-architecture protocol. For each task we instantiate one architecture and two variants differing *only* in the hidden nonlinearity: a conventional unit (ReLU for the convolutional perception net; tanh for the recurrent LM) and a leaky-integrate-and-fire (LIF) spiking unit with surrogate-gradient training. All weight shapes,

depths, embeddings and readouts are identical, so any energy gap is attributable to spiking alone.

Attainable-sparsity probe. We add a two-sided target-rate regularizer $\mathcal{L}_{\text{reg}} = \sum_{\ell} |\bar{s}_{\ell} - \rho^*|$ with warmup, penalizing deviation from a target rate ρ^* in *both* directions. Unlike one-sided L_1 (which collapses the network into a silent, unrecoverable state), this lets us *request* a firing rate and observe whether the task permits it.

Energy proxy. Per 45 nm estimates [5], $\text{MAC} = 4.6 \text{ pJ}$, $\text{AC} = 0.9 \text{ pJ}$. Dense transforms fed by continuous signals count as MACs; spike-driven transforms count as ACs scaled by measured presynaptic firing and timesteps T . We separate the *sparsity-driven* component from the fixed *per-op* ($\text{AC} < \text{MAC}$) component.

3 The sparsity ceiling: a firing-floor bound

Consider a spiking recurrent network with H hidden units communicating by binary spikes $s_t \in \{0, 1\}^H$, whose recurrent and readout pathways depend on the state *only through* s_t —the operative regime on spike-based neuromorphic hardware. Let $\rho = \mathbb{E}[\bar{s}_t]$.

Proposition 1 (Firing floor from memory load)

If the task requires distinguishing among M memory contents per recall step (copying N symbols from an alphabet of size S gives $M = S^N$), then, taking the branch $\rho \leq 1/2$,

$$\rho \geq H_b^{-1}\left(\frac{\log_2 M}{H}\right), \quad H_b(\rho) = -\rho \log_2 \rho - (1-\rho) \log_2 (1-\rho).$$

Hence $\rho_{\min}$ is strictly increasing in the memory load $\log_2 M = N \log_2 S$, and $\rho_{\min} \rightarrow 0$ only as $\log_2 M/H \rightarrow 0$.

Proof sketch. The spike vector s_t is the sole information channel out of the state. A binary vector of length H with at most $k = \rho H$ ones indexes at most $\sum_{i \leq k} \binom{H}{i} \leq 2^{HH_b(\rho)}$ messages ($\rho \leq 1/2$). Separating M contents requires $2^{HH_b(\rho)} \geq M$, i.e. $HH_b(\rho) \geq \log_2 M$. $\square$

Corollary 1 (Load, not memory, sets the floor) *If the task emits one decision among C classes (not maintained across time), the constraint is $HH_b(\rho) \geq \log_2 C$ at the readout only, so ρ can approach a far smaller minimum when $C \ll M$. Memory is thus not privileged per se: any source of representational load raises the floor. What makes sequence modeling special is that memory makes $M=S^N$ grow exponentially in the horizon, whereas typical (low-class-count) perception carries a modest M and sparsifies freely.*

§5 confirms both halves: a memoryless task’s floor still rises with C , yet the exponential memory term is what

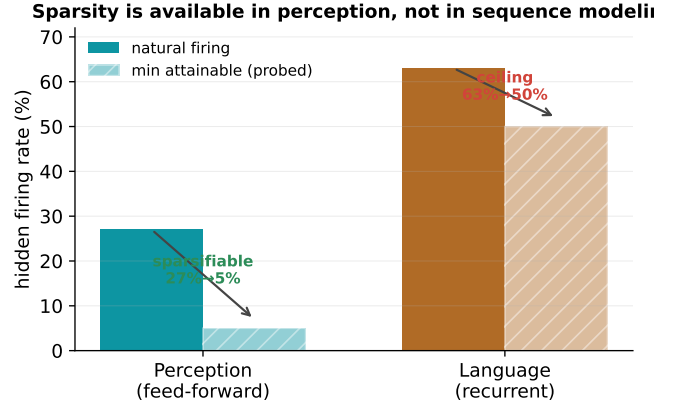


Figure 1: The paper in one plot. The two-sided probe drives perception firing from 27% to 5% with no accuracy loss, but cannot push recurrent language below $\sim 50\%$: sparsity is available in feed-forward perception and structurally unavailable in recurrent sequence modeling.

floors recurrent language modeling. *Scope:* the bound assumes spike-only state communication; reading the continuous membrane potential loosens it, but forfeits the spike-based communication that defines neuromorphic hardware.

4 Experiments

Results are mean $\pm$ std over 3 seeds unless noted.

4.1 Perception is freely sparsifiable

Convolutional SNN vs. matched CNN (rate-coded FashionMNIST as an event-stream stand-in; native event data is future work). Firing drives precisely to the 5% target with accuracy on par with the unregularized SNN; one-sided L_1 collapses the network.

config	firing	acc	ops/ANN	energy/ANN
ANN (CNN)	—	88.1%	1.0×	1.0×
SNN, no reg	.16/.27	74.7%	0.52×	2.7×
SNN, L_1	0/0	10%	—	collapse
SNN, $\rho^*=0.05$	.050/.051	76.6 $\pm$.4	0.99×	5.05×

4.2 Sequence modeling hits a ceiling

Spiking recurrent LM vs. architecturally identical tanh RNN (char-level WikiText-103). A strong penalty targeting 10% moves firing only to 0.50 (from 0.63), tight across seeds; the quality gap exceeds perception’s, and the SNN trains $\sim 6\times$ slower on GPU.

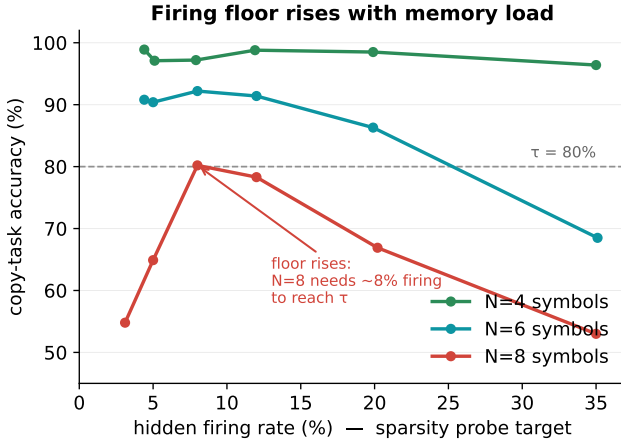


Figure 2: Copy-task accuracy vs. hidden firing rate for three memory loads N . The firing needed to reach the accuracy threshold rises with N : the sparsity floor tracks memory demand, as predicted by Prop. 1.

config	firing	bpc	energy/ANN
ANN (tanh RNN)	—	2.62	1.0×
SNN, no reg	$0.63 \pm .03$	$3.45 \pm .03$	4.6×
SNN, $\rho^*=0.10$	$0.50 \pm .01$	$3.48 \pm .02$	5.1×

4.3 The firing floor rises with memory load

On a synthetic copy task (ANN solves all, acc 1.00), we trace accuracy vs. attained firing for memory loads $N \in \{4, 6, 8\}$ (Fig. 2). At $N=4$ the task is solved at the lowest probed firing ($\sim 4\%$); at $N=8$ accuracy collapses below $\sim 8\%$ firing and only recovers there. The minimum firing that sustains accuracy **rises** ($\sim 4\% \rightarrow \sim 8\%$), corroborating the bound. Accuracy is also *non-monotone* in firing—excess spikes hurt—indicating an optimal mid-low band.

4.4 The input floor

Layer-wise op accounting at $\rho^*=0.05$ on perception: even with hidden layers at 5% firing, the input layer—a dense frame replayed over $T=10$ steps—is 61% of the op budget, capping op reduction at parity ($0.99\times$). This isolates *native event-driven input* (sparse by construction) as the missing ingredient for a true op reduction, not merely an energy-per-op reduction.

4.5 The floor scales with representational load (N , H , and C)

The bound depends on $\log_2 M/H$, where M is the number of messages the spike code must separate. Two controlled sweeps test its structure.

Memory load and width (Fig. 4). On the copy task ($M=S^N$) we sweep memory load N and hidden width H

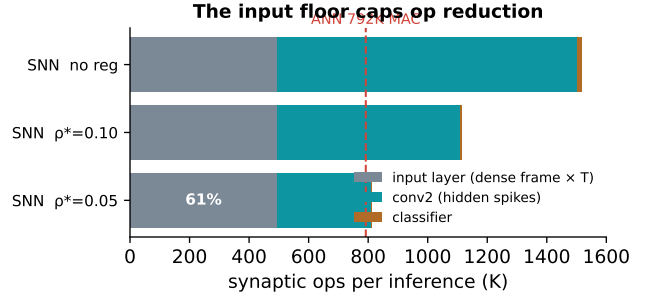


Figure 3: Layer-wise synaptic-op budget. As hidden firing is driven down (ρ^* : none $\rightarrow 0.10 \rightarrow 0.05$), the dense input layer (replayed over T) becomes the dominant cost (61% at $\rho^*=0.05$), holding total ops at parity with the ANN’s MAC count. Native event-driven input would remove this floor.

and read accuracy at a fixed 5% firing budget. Accuracy falls as N grows (the floor rises) but is bought back by larger H (at $N=8$, $H=128$ collapses to 48% while $H=512$ holds 77%) — both directions match the $\log_2 M/H$ structure of the bound.

Difficulty (Fig. 5). We had conjectured the floor tracks *memory* specifically. It does not: a feed-forward classification with *no* memory but growing class count C also raises the floor ($9.5\% \rightarrow 18\% \rightarrow 26\%$ for $C=10, 40, 160$). This *refutes* a clean memory-vs-difficulty dissociation but *confirms the bound’s generality*: any source of representational load ($M=C$ here) raises $\rho_{\min}$. The reason sequence modeling is uniquely floored is then sharp: memory makes $M=S^N$ grow *exponentially* in the horizon, whereas typical perception carries a modest M . The vision–language gap is memory’s exponential blow-up of M , not a categorical memory-only effect — and “perception is freely sparsifiable” holds only at *low representational load*.

4.6 Attention sidesteps the ceiling—but pays a memory wall

Is the ceiling intrinsic to sequence modeling, or to *recurrence*? We build a spiking Transformer (spiking attention-output and FFN neurons, integrated over $T_s=4$ spike-steps) and its matched ANN Transformer on the same char-level task, and apply the probe (3 seeds).

config	firing	bpc
ANN Transformer	—	$3.05 \pm .01$
SNN, no reg	$0.27 \pm .01$	$3.55 \pm .01$
SNN, $\rho^*=0.05$	$0.050 \pm .001$	3.57
SNN, $\rho^*=0.02$	$0.020 \pm .000$	3.57

Firing drops $14\times$ ($0.27 \rightarrow 0.02$) with *no* bpc change: the spiking Transformer sparsifies as freely as feed-forward perception, and shows *none* of the recurrent floor (Fig. 6).

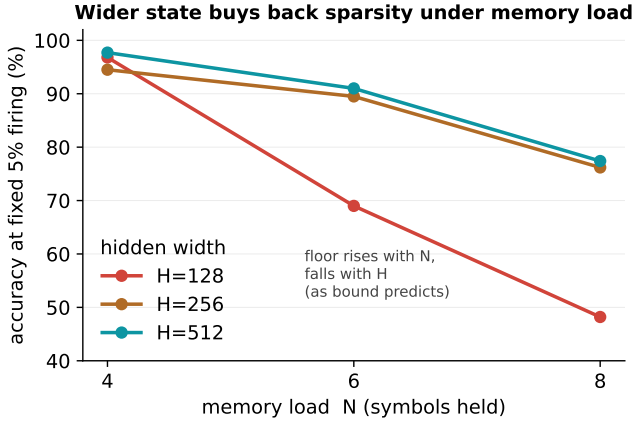


Figure 4: Accuracy at a fixed 5% firing budget vs. memory load N , for three hidden widths H . The sparsity floor rises with N and falls with H , as $\rho_{\min} \approx H_b^{-1}(\log_2 M/H)$ predicts.

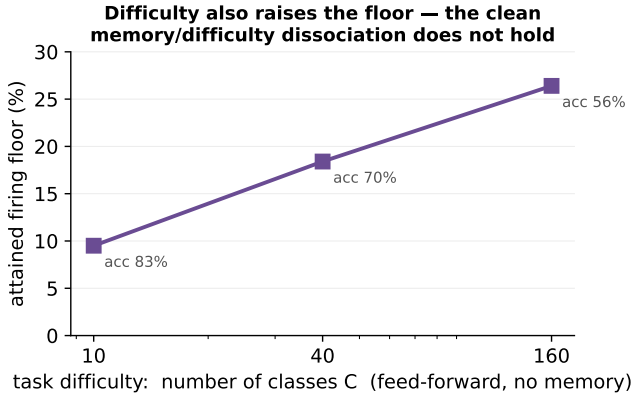


Figure 5: A memoryless feed-forward task’s firing floor still rises with difficulty C : representational load, not memory alone, sets the floor—consistent with the bound ($M=C$).

The ceiling is thus a property of *recurrent compression*—forcing history through a fixed-width spike state that must stay active—not of sequence modeling per se. Attention avoids it by keeping all tokens’ key-value entries (random access, no compression), i.e. it trades the firing floor for $O(\text{context})$ memory—the on-chip memory wall that blocks large models on neuromorphic silicon [4]. On sequence tasks, neuromorphic hardware therefore faces a dichotomy—firing floor (recurrent) or memory wall (attention)—and escapes neither.

5 Implications and related work

Where neuromorphic wins: sparse, event-driven perception—low attainable firing and, with event sensors, no input floor. **Where it does not:** sequence models

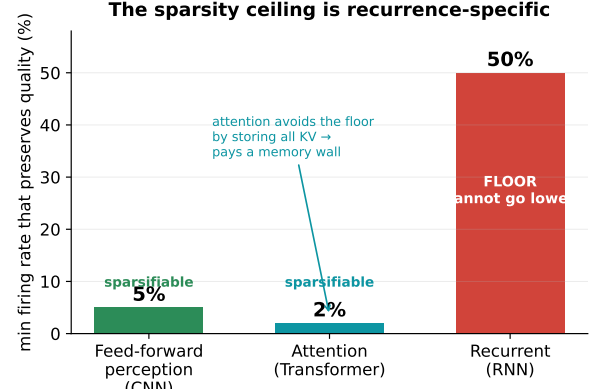


Figure 6: Minimum firing that preserves quality, by architecture. Feed-forward perception (5%) and attention (2%) sparsify freely; the recurrent net is floored near 50%. Attention buys its sparsity with a KV-cache memory wall.

face a dichotomy. A *recurrent* SNN compresses history into a fixed spike state and hits the firing floor (no sparsity dividend, only the fixed $\text{AC} < \text{MAC}$ constant that efficient ANNs also chase). An *attention* SNN sparsifies freely but stores the full key-value cache— $O(\text{context})$ memory—colliding with the on-chip-memory wall [4] that already blocks LLM-scale weights on a neuromorphic chip. Either way the cost re-appears; it is not removed. Consistently, the strongest “spiking LLM” results (SpikingBrain [2], 76B) run on GPU clusters and spike Transformers [3] largely target vision. High firing is known to erase SNN efficiency [6, 7]; we add the controlled-probe cross-task view, a bound, and the recurrence-vs-attention cost dichotomy.

6 Limitations

- (1) Perception uses rate-coded FashionMNIST as an event-stream stand-in; native N-MNIST/DVS and a real event-input accounting are needed.
- (2) The spiking Transformer is single-layer, char-level, small-scale; scaling it (and quantifying the KV-cache memory cost directly) is the key next step.
- (3) The bound is single-step; a tightening across T and to task loss (not just distinguishability) is open.
- (4) Energy is a 45 nm proxy; measured Loihi 2 / SpiNNaker2 energy would convert estimate to measurement.
- (5) The mechanism sweep uses a narrow trainable window ($N \leq 8$).

Code and reproducibility

All scripts, per-run logs, and the energy-model constants are released; every result runs on a single GPU with no custom hardware. *Repository:* github.com/zeyuyuyu/sparsity-ceiling (made public at submission).

References

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